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ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 17-03-2026**

Time: 09.00Am to 12.00Pm **GTM-23** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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17-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-23_Test Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let the position vectors of the points A, B, C and D $5\hat{i}+5\hat{j}+2\lambda\hat{k}, \hat{i}+2\hat{j}+3\hat{k}, -2\hat{i}+\lambda\hat{j}+4\hat{k}$ and $-\hat{i}+5\hat{j}+6\hat{k}$. Let the set $s=\{\lambda\in R:$ the point A, B, C and D are coplanar$\}$. Then $\sum_{\lambda\in s}(\lambda+3)^2$ is equal to

1) $\frac{37}{2}$ **2)** 13 **3)** 61 **4)** 41

2. A pair of fair dice is throw independently three times. The probability of getting a score of exactly 9 twice is

1) $\frac{8}{243}$ **2)** $\frac{1}{729}$ **3)** $\frac{8}{9}$ **4)** $\frac{8}{729}$

3. Let $A=\left[a_{ij}\right]_{2x2}$ where $a_{ij}\neq 0$ for all i, j and $A^2=I$. Let 'a' be the sum of all diagonal elements of A and $b=|A|$. Then $3a^3+5b^4$ is equal to:

1) 5 **2)** 4 **3)** 3 **4)** 7

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4. If the coefficients of three consecutive terms in the expansion of $(1+x)^n (n\in N)$ are in the ratio 1:5:20 then the coefficient of fourth term is

1) 5481 **2)** 3654 **3)** 2436 **4)** 1817

5. Let A, B, C be 3×3 matrices such that A is symmetric and B and C are Skew –symmetric consider the statements.

$(s_1)A^{31}B^{52}-B^{52}A^{31}$ is symmetric

$(s_2)A^{52}C^{31}-C^{31}A^{52}$ is symmetric then

1) Only s_2 is true **2)** Both s_1 and s_2 are false

3) Both s_1 and s_2 are true **4)** only s_1 is true

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6. The value of $54\left(4\cos^2 9^0-1\right)\left(4\cos^2 27^0-1\right)\left(4\cos^2 81^0-1\right)\left(4\cos^2 243^0-1\right)$ is

1) 54 **2)** 18 **3)** 27 **4)** 36

7. Statement-1 : $11^{25}+12^{25}$ when divided by 23 leaves the reminder zero.

Statement-2 : a^n+b^n is always divisible by $(a+b)\forall a,b.n\in N$

1) Both statement – 1 and statement – 2 are true

2) Both statement– 1 and statement – 2 are false

3) Statement – 1 is true; statement – 2 is false

4) Statement – 1 is false; statement – 2 is true

8. The domain of the function

$f(x)=\dfrac{1}{\sqrt{[x]^2-3[x]-18}}$ is (where [.] is G.I.F)

1) $(-\infty,-4)\cup(7,\infty)$ **2)** $(-\infty,-3)\cup[7,\infty)$ **3)** $(-\infty,-4)\cup(6,\infty)$ **4)** $(-\infty,4)\cup(7,\infty)$

9. For two non-zero complex numbers z_1 and z_2 if $\mathrm{Re}(z_1z_2)=0$ and $\mathrm{Re}(z_1+z_2)=0$ then

A) $\mathrm{Im}(z_1)>0 and\ \mathrm{Im}(z_2)>0$ B) $\mathrm{Im}(z_1)<0 and\ \mathrm{Im}(z_2)>0$

C) $\mathrm{Im}(z_1)>0 and\ \mathrm{Im}(z_2)<0$ D) $\mathrm{Im}(z_1)<0 and\ \mathrm{Im}(z_2)<0$

Choose the correct answer from the option given below

1) B and D **2)** B and C **3)**A and B **4)** A and C

10. The set of values of a for which all extremum of function $f(x)=x^3-3ax^2+3\left(a^2-1\right)x+1$ lie in the interval (-2,4) is

1) (3, 4) **2)** (-1, 3) **3)** (-3, -1) **4)** (-2, -1)

11. If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers then the serial number of the word THAMS is

1) 103 **2)** 104 **3)** 101 **4)** 102

12. The range of the function $f(x)=\sqrt{\cos^{-1}\left(\sqrt{\log_4^x}\right)-\dfrac{\pi}{2}+\sin^{-1}\left(\dfrac{1+x^2}{4x}\right)}$ is equal to

1) $\left(0,\dfrac{\pi}{2}+\sqrt{\dfrac{\pi}{2}}\right)$ **2)** $\left[\dfrac{\pi}{2},\dfrac{\pi}{2}+\sqrt{\dfrac{\pi}{2}}\right]$ **3)** $\left[\dfrac{\pi}{6},\dfrac{\pi}{2}\right]$ **4)** $\left\{\dfrac{\pi}{6}\right\}$

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13. The differential equation whose general solution is given by $y = C_1\cos(x+c_2) - C_3e^{-x+c_4} + C_5\sin x$ where $C_1, C_2, ----C_5$ are constants is

1) $\frac{d^4y}{dx^4} - \frac{d^2y}{dx^2} + y = 0$ **2)** $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$

3) $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ **4)** $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

14. Let $\alpha \in R$ be such that the function

$$f(x) = \begin{cases} \dfrac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}-\{x\}^3} & ;x \neq 0 \\ \alpha & x = 0 \end{cases}$$ is continuous at $x = 0$; where $\{x\} = x - [x]; [x]$ is the greatest integer less than or equal to x then:

1) $\alpha = 0$ **2)** no such α exists **3)** $\alpha = \frac{\pi}{4}$ **4)** $\alpha = \frac{\pi}{\sqrt{2}}$

15. $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to

1) 2π **2)** $\frac{\pi}{2}$ **3)** $\frac{\pi}{3}$ **4)** $\frac{\pi}{6}$

16. The maximum value of $f(x) = \begin{vmatrix} \sin^2 x & 1+\cos^2 x & \cos 2x \\ 1+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}; x \in R$ *is*

1) 5 **2)** $\sqrt{7}$ **3)** $\frac{3}{4}$ **4)** $\sqrt{5}$

17. Let α, β be two roots of the equation $x^2 + (20)^{\frac{1}{4}}x + (5)^{\frac{1}{2}} = 0$. Then $\alpha^8 + \beta^8$ is equal to

1) 10 **2)** 50 **3)** 5 **4)** 30

18. The coordinates of one of the point on the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$ which is at a distance of 3 units from the point (1, -2, 3) is

1) $\left(\frac{13}{7}, \frac{23}{7}, \frac{-3}{7}\right)$ **2)** $\left(\frac{13}{7}, \frac{-5}{7}, \frac{39}{7}\right)$ **3)** $\left(\frac{-1}{7}, \frac{5}{7}, \frac{-3}{7}\right)$ **4)** $\left(\frac{-1}{7}, \frac{-5}{7}, \frac{39}{7}\right)$

19. Let $y = y(x)$ be the solution of differential equation

$\frac{dy}{dx} = 1 + xe^{y-x}; -\sqrt{2} < x < \sqrt{2}; y(0) = 0$ then the minimum value of $y(x); x \in \left(-\sqrt{2}, \sqrt{2}\right)$ is equal to

1) $\left(2-\sqrt{3}\right) - \log_e^2$ **2)** $\left(1-\sqrt{3}\right) - \log_e^{\left(\sqrt{3}-1\right)}$ **3)** $\left(1+\sqrt{3}\right) - \log_e^{\left(\sqrt{3}-1\right)}$ **4)** $\left(2+\sqrt{3}\right) + \log_e^2$

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20. **Statement I:** The point A(1, 0, 7) is the mirror image of the point B(1, 6, 3) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$

Statement II: The line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ bisects the line segment joining A(1, 0, 7) and B(1, 6, 3)

1) Statement -I false, Statement -II is true

2) Statement -I is false; Statement -II is false

3) Statement -I is true, Statement -II is true

4) Statement – I is true; Statement – II is false

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If the area of the region bounded by the curves $y=ex\log x$ and $y=\frac{\log x}{ex}$ is $\frac{k}{4e}$, then the value of [k] is_____ (where [.] represents greatest integer function)

22. For a unique value of $\mu\,\&\,\lambda$ the system of equations given by

$x+y+z=6$

$x+2y+3z=14$

$2x+5y+\lambda z=\mu$

Has infinitely many solutions then $\frac{\mu-\lambda}{4}$ is equal to

23. If the coefficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ and the coefficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to

24. $e_1\,\&\,e_2$ are eccentricities of $\frac{x^2}{18}+\frac{y^2}{4}=1$ and $\frac{x^2}{9}-\frac{y^2}{4}=1$ respectively. If (e_1,e_2) lies on $15x^2+3y^2=k$, the value of $\frac{k}{2}=$ _____

25. The shortest distance between the lines $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-6}{2}$ and $\frac{x-6}{3}=\frac{1-y}{2}=\frac{z+8}{0}$ is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

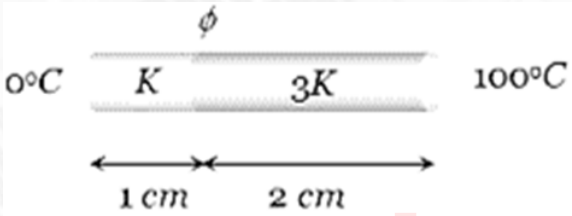
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. A bus is moving with a velocity of 10 m/s on a straight road. A scooterist wishes to overtake the bus in 100 seconds. If the bus is at a distance of 1 km form the scooterist, at what velocity should the scooterist chase the bus:

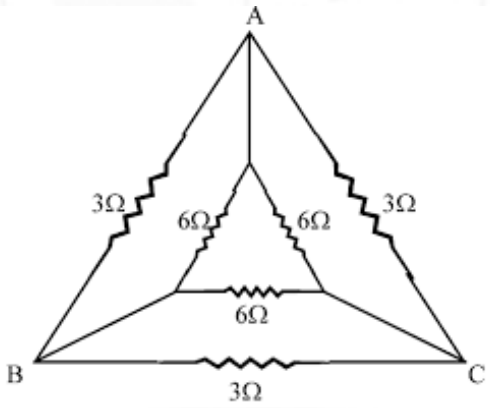
1) 50 m/sec **2)** 40 m/sec **3)** 30 m/sec **4)** 20 m/sec

27. Two bars of thermal conductivities K and 3K and lengths 1 cm and 2 cm, respectively, have equal cross-sectional area. They are joined length wise as shown in the figure. If the temperature at the ends of this composite bar is $0^{\circ}C$ and $100^{0}C$, respectively (see figure), then the temperature ϕ of the interface is:



1) $50^{0}C$ **2)** $\frac{100}{3^{0}C}$ **3)** $60^{0}C$ **4)** $\frac{200}{3^{0}C}$

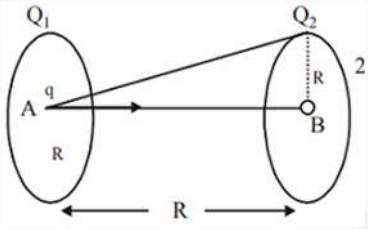
28. In the circuit shown the effective resistance between B and C is:



1) 3Ω **2)** 4Ω **3)** $\frac{4}{3}\Omega$ **4)** $\frac{3}{4}\Omega$

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29. Two identical thin rings, each of radius R metres, are coaxially placed at a distance R metres apart. If Q_1 coulomb and Q_2 coulomb are respectively, the charges uniformly spread on the two rings, the work done in moving a charge q from the centre of one ring to that of the other is:



1) Zero

2) $\frac{q(Q_1-Q_2)\left(\sqrt{2}-1\right)}{\sqrt{2}4\pi\varepsilon_0 R}$

3) $\frac{q\sqrt{2}(Q_1+Q_2)}{4\pi\varepsilon_0 R}$

4) $\frac{q(Q_1+Q_2)\left(\sqrt{2}+1\right)}{\sqrt{2}4\pi\varepsilon_0 R}$

30. Time period of a simple pendulum is T inside a lift when the lift is stationary. If the lift moves upwards with an acceleration g/2, the time period of pendulum will be:

1) $\sqrt{\frac{3}{2}}T$ **2)** $\frac{T}{\sqrt{3}}$ **3)** $\sqrt{3}T$ **4)** $\sqrt{\frac{2}{3}}T$

31. A uniform thin rod AB of length L has linear mass density $\mu(x)=a+\frac{bx}{L}$, where x is measured from A. If the CM of the rod lies at a distance of $\left(\frac{7}{12}\right)L$ from A, then a and b are related as :

1) $a=2b$ **2)** $2a=b$ **3)** $a=b$ **4)** $3a=2b$

32. The ratio of specific heats $\left(\frac{C_P}{C_V}=\gamma\right)$ in terms of degree of freedom (f) is

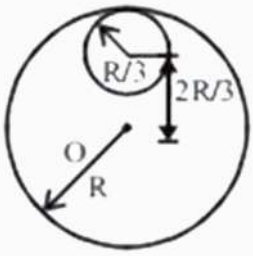
(C_P = Specific heat capacity at constant Pressure

C_V = Specific heat capacity at constant Volume)

1) $1+\frac{f}{3}$ **2)** $1+\frac{2}{f}$ **3)** $1+\frac{f}{2}$ **4)** $1+\frac{1}{f}$

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33. A disc has mass 9m and radius R. A hole of radius R/3 is cut from it as shown in the figure. The moment of inertia of remaining part about an axis passing through the centre 'O' of the disc and perpendicular to the plane of the disc is:



1) $8mR^2$ **2)** $4mR^2$ **3)** $\frac{40}{9}mR^2$ **4)** $\frac{37}{9}mR^2$

34. In a standard YDSE, the distance between the slits varies with time t as $d(t) = d_0 + a_0\sin(\omega t)$ .The difference between the largest fringe width and the smallest fringe width over time is given as (λ is wavelength of monochromatic light used, D is the screen distance from slits. D >>d)

1) $\frac{2\lambda D a_0}{d_0^2 - a_0^2}$ **2)** $\frac{2\lambda D d_0}{d_0^2 - a_0^2}$ **3)** $\frac{\lambda D}{d_0 + a_0}$ **4)** $\frac{\lambda D a_0}{d_0^2 - a_0^2}$

35. The magnetic field of a plane electromagnetic wave is given by $B = 3\times10^{-8}\left[\sin\left(200\pi\left(y+ct\right)\right)\right]\hat{i}T$. Where $C = 3\times10^8$ m/s is the speed of light. The corresponding electric field is

1) $\bar{E} = 9\sin\left[\left(200\pi\left(y+ct\right)\right)\right]\hat{k}\,v/m$ **2)** $\bar{E} = -10^{-6}\sin\left[\sin\left(200\pi\right)\left(y+ct\right)\right]\hat{k}\,v/m$

3) $\bar{E} = 3\times10^{-8}\left[\sin\left(200\pi\left(y+ct\right)\right)\right]\hat{k}\,v/\bar{m}$ **4)** $\bar{E} = -9\sin\left[\left(200\pi\left(y+ct\right)\right)\right]\hat{k}\,v/m$

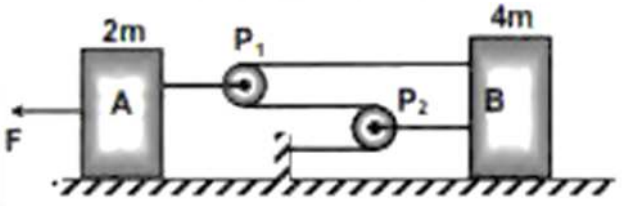
36. The relation between internal energy U , pressure P and volume V of a gas in an adiabatic process is $U = a + bPV$.

Where a and b are constant. What is the effective value of adiabatic constant γ ?

1) $\frac{a}{b}$ **2)** $\frac{b+1}{b}$ **3)** $\frac{a+1}{a}$ **4)** $\frac{b}{a}$

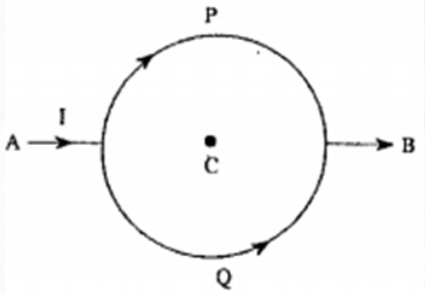
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37. The acceleration of the block B shown in the figure will be (Assuming the surfaces and the light pulleys P_1 and P_2 all are smooth)



1) $\frac{F}{4m}$ **2)** $\frac{F}{6m}$ **3)** $\frac{F}{2m}$ **4)** $\frac{3F}{17m}$

38. Assertion (A): The magnetic field at the center of the current carrying circular coil shown in the fig. is zero.



Reason (R): The magnitudes of magnetic fields are equal and the directions of magnetic fields due to both the semicircles are opposite.

1) Both A and R are true but R is the correct explanation of A
2) Both A and R are true but R is not the correct explanation of A
3) A is true but R is false
4) A is false but R is also false

39. Match the List-I with the List-II.

List-I		List-II	
A.	Phase difference between current and voltage in a purely resistive AC circuit	1.	$\frac{\pi}{2}$; current leads voltage
B.	Phase difference between current and voltage in a pure inductive AC circuit	2.	Zero
C.	Phase difference between current and voltage in a pure capacitive AC circuit	3.	$\frac{\pi}{2}$; current lags voltage
D.	Phase difference between current and voltage in an LCR series circuit	4.	$\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

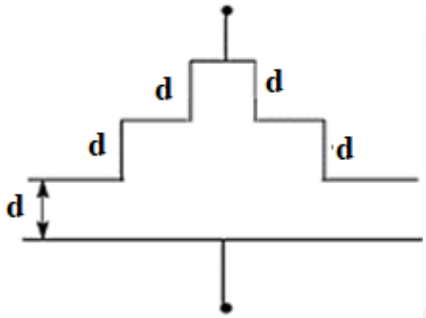
Choose the most appropriate answer from the options given below:

1) A-2, B-3, C-1, D-4 **2)** A-1, B-3, C-4, D-2
3) A-2, B-3, C-4, D-1 **4)** A-2, B-4, C-3, D-1

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40. The upper plate of parallel plate capacitor of plate area A is modified into 5 equal segments as shown. The equivalent capacitance between the terminals is __



1) $\frac{10\epsilon_0 A}{3d}$ **2)** $\frac{2\epsilon_0 A}{3d}$ **3)** $\frac{3\epsilon_0 A}{10d}$ **4)** $\frac{3\epsilon_0 A}{2d}$

41. If a string of length l fixed at both ends vibrates with a standing wave $y = A\sin\left(\frac{2\pi}{l}X\right)\sin 2\pi t$ in resonance. Then the minimum time (from t=0) after which energy is maximum at mid-point of string will be_____

1) $\frac{1}{4}$ sec **2)** 5 sec **3)** $\frac{1}{8}$ sec **4)** $\frac{1}{6}$ sec

42. Match the entries in the Column I with those in Column II

Column I		Column II	
(A)	Binding energy per nucleon of a nucleus	(P)	Photoelectric effect
(B)	Particle nature of light	(Q)	Nuclear fission and fusion
(C)	Binding energy of products is greater than that of reactants	(R)	Uncontrolled chain reaction
(D)	Atom bomb	(S)	Measure of stability

1) A – S; B – P; C – R; D – Q **2)** A – S; B – P; C – Q; D - R

3) A – P; B – S; C – Q; D – R **4)** A – S; B – Q; C – P; D – R

43. **STATEMENT – 1:** Kinetic energy of a particle is conserved if it is acted upon by a conservative force only.

STATEMENT – 2: Work done by a conservative force in a closed path is zero.

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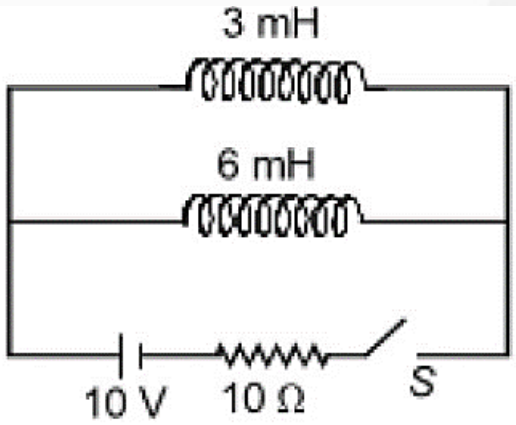
1) Statement – 1 is True, Statement – 2 is True; Statement – 2 **is** a correct explanation for Statement – 1.

2) Statement – 1 is True, Statement – 2 is True; Statement – 2 **is NOT** a correct explanation for Statement – 1.

3) Statement – 1 is True, Statement – 2 is False.

4) Statement – 1 is False, Statement –2 is True.

44. The current through 3 mH inductor in steady state after closing switch S is



1) $\frac{1}{3}$ ampere **2)** $\frac{2}{3}$ ampere **3)** 1 ampere **4)** $\frac{3}{2}$ ampere

45. A horizontal overhead power line is at a height of 4 m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is $\left(\mu_0 = 4\pi \times 10^{-7} TmA^{-1}\right)$

1) $2.5\times10^{-7}T$ northwards **2)** $2.5\times10^{-7}T$ southwards

3) $5\times10^{-6}T$ northwards **4)** $5\times10^{-6}T$ southwards

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

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46. An artificial satellite is moving in a circular orbit around the earth with a speed equal to half of the magnitude of escape velocity from the earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely onto the earth, then speed $\left(in\, ms^{-1}\right)$ with which it hits the surface of the earth is $\sqrt{kgR}$. Find the value of k.

$\left(Given = 9.8ms^{-2} and\ R = 6.4\times10^{6}\, m\right)$.

47. A prism of refractive index μ and angle of prism A is placed in position of minimum angle of deviation. If minimum angle of deviation is also A, then in terms of refractive index the value of A is $\alpha\cos^{-1}\left(\frac{\mu}{\beta}\right)$. The value of $\alpha+\beta$ is _____.

48. An α - particle and a proton are accelerated from rest by a potential difference of 100 v. After this their de Broglie's wavelength are λ_α and λ_P . The ratio $\frac{\lambda_P}{\lambda_\alpha}$ to nearest integer is_____.

49. Pitch of a screw gauge is 1mm and its cap is divided into 100 divisions. When nothing is placed between studs of the screw gauge, zero of circular scale is 8 divisions above the reference line and zero of the main scale is not visible. Now, when a cylindrical wire is placed between its studs the main scale reading is 2 divisions and 15th division of circular scale coincides with reference line. Diameter of the wire is x mm.
Find the value of 100 x.

50. M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is _____ grams.

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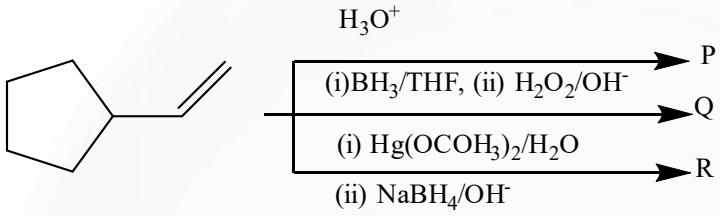
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Which is correct for product, P, Q and R (P, Q, R are major product)



1) Products P & R are identical

2) Products Q & R are identical

3) Products P & Q are functional group isomers

4) Products Q & R are position isomers

52. Which of the following statement is incorrect?

1) SRP values of halogens $X_2(g)/X^-_{(aq)}$ $F_2 > Cl_2 > Br_2 > I_2$

2) Bond dissociation enthalpy of $Cl_2 > F_2 > Br_2 > I_2$

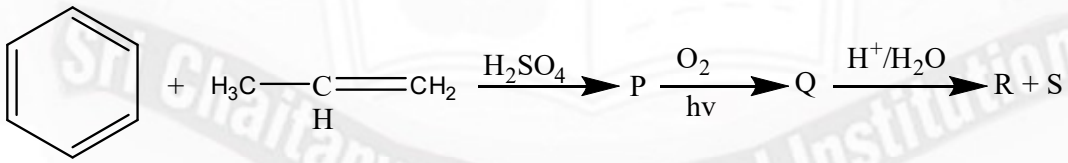
3) Boiling points of $I_2 > Br_2 > Cl_2 > F_2$

4) Reducing power of $I^- > Br^- > Cl^- > F^-$

53. An electron in an atom jumps to the higher energy level in such a way that its kinetic energy changes from 'y' to $\frac{y}{2}$. Then change in its potential energy will be

1) y **2)** $-y$ **3)** $+\frac{y}{2}$ **4)** $-\frac{y}{2}$

54.



If R is aromatic and S is aliphatic, then:

I) Rate of EAS of (R) is more than that in benzene.

II) Enol content of "S" is more than the enol content of acetaldehyde

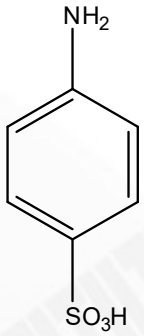
III) R is more acidic than C_2H_5OH

1) I and III are correct **2)** II and III are correct

3) Only III is correct **4)** All I, II and III are correct

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55. Which of the following will not give Lassaigne's test for N in sodium extract?



1) $C_6H_5NHNH_2$ **2)** NH_2CONH_2 **3)** NH_2-NH_2 **4)**

56. The compound which has zero dipole moment is

1) CH_2Cl_2 **2)** cis But – 2 – ene

3) PCl_3F_2 **4)** Ortho-Dichlorobenzene

57. Cerium (Z = 58) is an important member of lanthanoids, which of the following statements about cerium is incorrect?

1) The common oxidation states for cerium are +3 and +4

2) The +3 oxidation state of cerium is more stable than +4

3) The +4 oxidation state of cerium is not known in solutions

4) Cerium (IV) acts as an oxidizing agent.

58. The complex, $\left[Co(NH_3)_4(NO_2)_2\right]Cl$ exhibits

1) Ionization isomerism, geometrical isomerism and optical isomerism

2) Linkage isomerism, geometrical isomerism and optical isomerism

3) Linkage isomerism, geometrical isomerism and ionization isomerism

4) Optical isomerism, ionization isomerism and Linkage isomerism

59. Choose the correct statement(s)

(A) If the solubility of Sb_2S_3 $1.0\times10^{-5}\,mol/L$ is at 298 K, its solubility product will be 108×10^{-25}

(B) $Keq\ of\ H_2O = 1.8\times10^{-16}$ at 25^0C

(C) Ammonia is a levelling solvent for stronger acids like HCl, HBr, HI while glacial acetic acid is differentiating solvent.

1) A and B **2)** B and C **3)** A and C **4)** A, B and C

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60. The osmotic pressure of blood at $37^{o}C$ is 8.21 atm. The amount of glucose (in gm) that should be added per litre for an intravenous injection so that it is isotonic with blood is (GMW of glucose = 180g and R = 0.082L atm $mol^{-1}K^{-1}$)

1) 20 gm **2)** 36 gm **3)** 42 gm **4)** 58 gm

61. Which of the following statement is correct for an aqueous solution of CH_3COOH with concentration $5\times10^{-2}M$ and having $Ka=2\times10^{-5}(\log 2=0.3)$

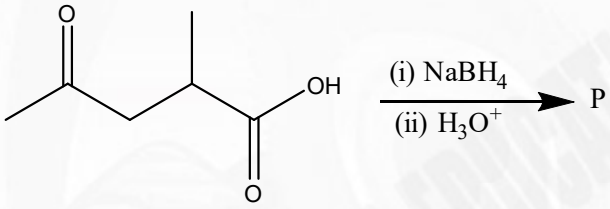
1) Its pH = 3.0

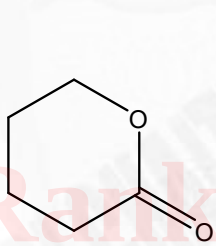
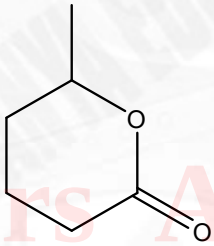
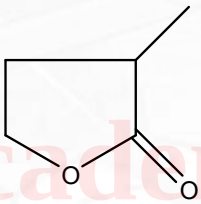
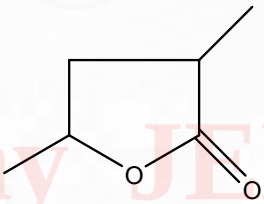
2) If equal moles of NaOH are added then pH = 7

3) It acts as acidic buffer if NaCl is added

4) It acts as basic buffer on adding NaOH

62. The product P in the following reaction is



1)  **2)**  **3)**  **4)** 

63. The mechanism of S_N1 reaction is given as

$$RX \rightarrow \underset{\text{ion pair}}{R^{\oplus}X^{\Theta}} \rightarrow \underset{\text{solvent Sepaerated ion pair}}{R^{\oplus} \parallel X^{\Theta}} \xrightarrow{Y^{\Theta}} RY + X^{\Theta}$$

A student writes general characteristics based on the given mechanism as

a) The reaction is favoured by strong nucleophiles

b) $R^{\oplus}$ Would be easily formed if the substituents are bulky

c) The reaction is accompanied by racemisation

d) The reaction is favoured by non-polar solvents

Which observations are correct?

1) a and b **2)** b and c **3)** a, b and c **4)** b and d

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64. The compound of xenon that has the same number of lone pairs as in I_3^- is (on central atom)

1) XeF_2 **2)** XeO_3 **3)** XeF_4 **4)** XeO_4

65. Assertion (A): Aniline on nitration gives meta nitro aniline in maximum yield than O, P isomers

Reason (R): $-NH_3^+$ acts as Meta directing group.

1) Both A and R are true R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A

3) A is true but R is false

4) A is false but R is true

66. According to MO theory which of the list ranks the oxygen species in terms of decreasing bond order $O_2, O_2^+, O_2^-, O_2^{2-}$

1) $O_2^{2-}, O_2^-, O_2^+, O_2$ **2)** $O_2^+, O_2, O_2^-, O_2^{2-}$

3) $O_2, O_2^+, O_2^-, O_2^{2-}$ **4)** $O_2^{2-}, O_2^-, O_2, O_2^+$

67. Which of the following statement is incorrect?

1) $E^0_{Agcl/Ag/cl-} = 0.24$ if $E^0_{Ag^+/Ag} = 0.84V$ and $K_{sp} AgCl = 10^{-10}$ $\left(\text{use} \frac{2.303RT}{F} = 0.06\right)$

2) λ_0^m for $H^+_{(aq)}$ is highest is aqueous solution

3) In the electrolysis of aqueous Na_2SO_4 if 11.2L of $H_{2(g)}$ is liberated at cathode, then at the anode the volume of $O_{2(g)}$ liberated is 22.4L at STP

4) In lead – acid battery the equivalent weight of $H_2SO_4 = 98$

68. In an atom, for a 3p-orbital there exist

1) Two spherical nodes
2) Two non-spherical nodes
3) One spherical and one non-spherical nodes
4) One spherical and two non-spherical nodes

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69. Which of the following is incorrect?

1) $\left[Fe(CN)_6\right]^{4-}$ and $\left[Fe(H_2O)_6\right]^{3+}$ have same number of unpaired e^- in central metal ion

2) A solution of $\left[Ni(H_2O)_6\right]^{2+}$ is green but a solution of $\left[Ni(CN)_6\right]^{2+}$ is colourless

3) $\left[Cr(NH_3)_6\right]^{3+}$ is paramagnetic while $\left[Ni(CN)_6\right]^{2-}$ is diamagnetic

4) d-orbital occupation of the central metal ion in the complex $\left[CoF_6\right]^{4-}$ is $t_{2g}^5 e_g^2$

70. Statement – I: Among 13th group elements, Gallium has maximum liquid range.
Statement – II: Oxidation of Tl in TlI_3 is + 3
Choose the correct option.

1) Both Statement – I and Statement – II are correct

2) Both Statement – I and Statement – II are incorrect

3) Statement – I is correct but Statement – II is incorrect

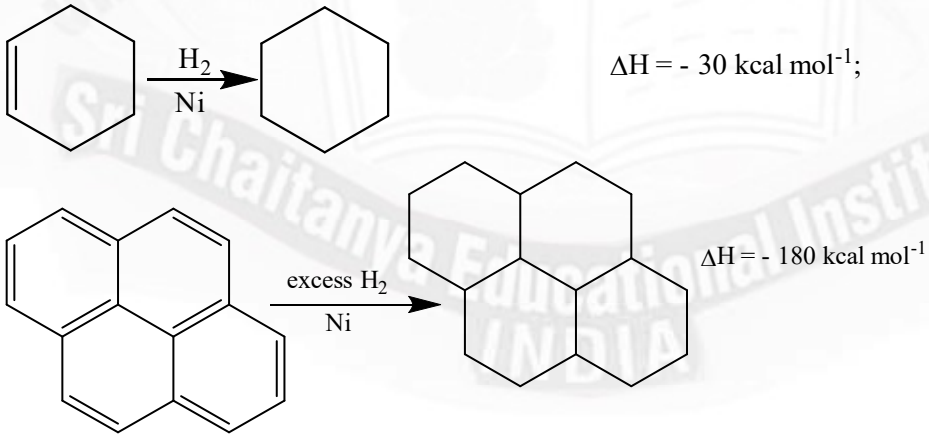
4) Statement – I is incorrect but Statement – II is correct

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Number of $-OH$ groups in one molecule of sucrose is……

72. Given that



Compound 'A'

What is the resonance energy of 'A' (in magnitude) is…..kcal mol-1

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73. For a first order reaction $A \rightarrow B$ the reaction rate at reactant concentration of 0.01M is found to be $3.0\times10^{-5}\, mol\ L^{-1}s^{-1}$. The half-life period of this reaction in seconds is

74. Identify the number of molecules with sp^3d hybridization on central atoms among the following.

$PCl_5, PO_4^{3-}, SF_4, XeF_2, I_3^-, XeO_2F_2, PCl_3F_2, SO_2Cl_2$

75. For the given reaction $A_{(g)} + 2B_{(g)} \rightleftharpoons 3C_{(g)}; K_c = 2\times10^{-3}$

Then $kp = x\times10^{-3}$, x is

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